

BMS Integration Specification Document

Smart Building Platform — Revised 2.1, March 15, 2026

1. Scope of Application

This document specifies the integration between **building management systems (BMS)** and a cloud platform. BMS is used to monitor and control HVAC systems, lighting, and electrical meters within a building.

2. System Architecture

- Air handling unit (AHU) controller, model **FX-PCG2611-0**
- Variable air volume (VAV) terminal controller
- Occupancy sensor and CO2 sensor (0–2,000 ppm)
- Edge device and gateway (BACnet/IP, Modbus TCP, MQTT)

3. Network Requirements

Item	Value
BMS server	192.168.1.100
BACnet/IP port	47808
MQTT broker	mqtt.example.com:8883
Polling interval	15 minutes

4. Alarm handling

When an alarm is triggered for the chiller, the system notifies the maintenance team via email (support@example.com) and generates a work order. If the alarm is not resolved within 30 minutes, it is escalated to the facility manager.

Note: Any change to the setpoint requires approval from the operator.

5. Energy Pipe

The (EMS) system records energy consumption every 15 minutes. During the summer of 2025, demand response and a change in the room setpoint from 24 °C to 26 °C helped reduce peak demand by 18%.

6. Data retention

The data from trend logs is stored on the cloud platform (<https://bms.example.com/trends>) for 13 months.

